ICU MORTALITY PREDICTION USING MACHINE LEARNING: A PRACTICAL FRAMEWORK FOR RISK STRATIFICATION

A Data Science Approach for Understanding and Identifying High-Risk Patients

Authors

Mashael Saeed, Sarah Elshiaty, Seifeldin Elshiaty

Instructor

Dr. Passent Elkafrawy

Introduction

This study analyzes and predicts ICU mortality using a data science framework. Early identification of high-risk patients is critical in intensive care settings. Traditional scoring systems like APACHE and SOFA provide useful indicators but may not capture complex clinical relationships. This project applies data-driven analysis to better understand patient risk and support improved clinical decision-making.

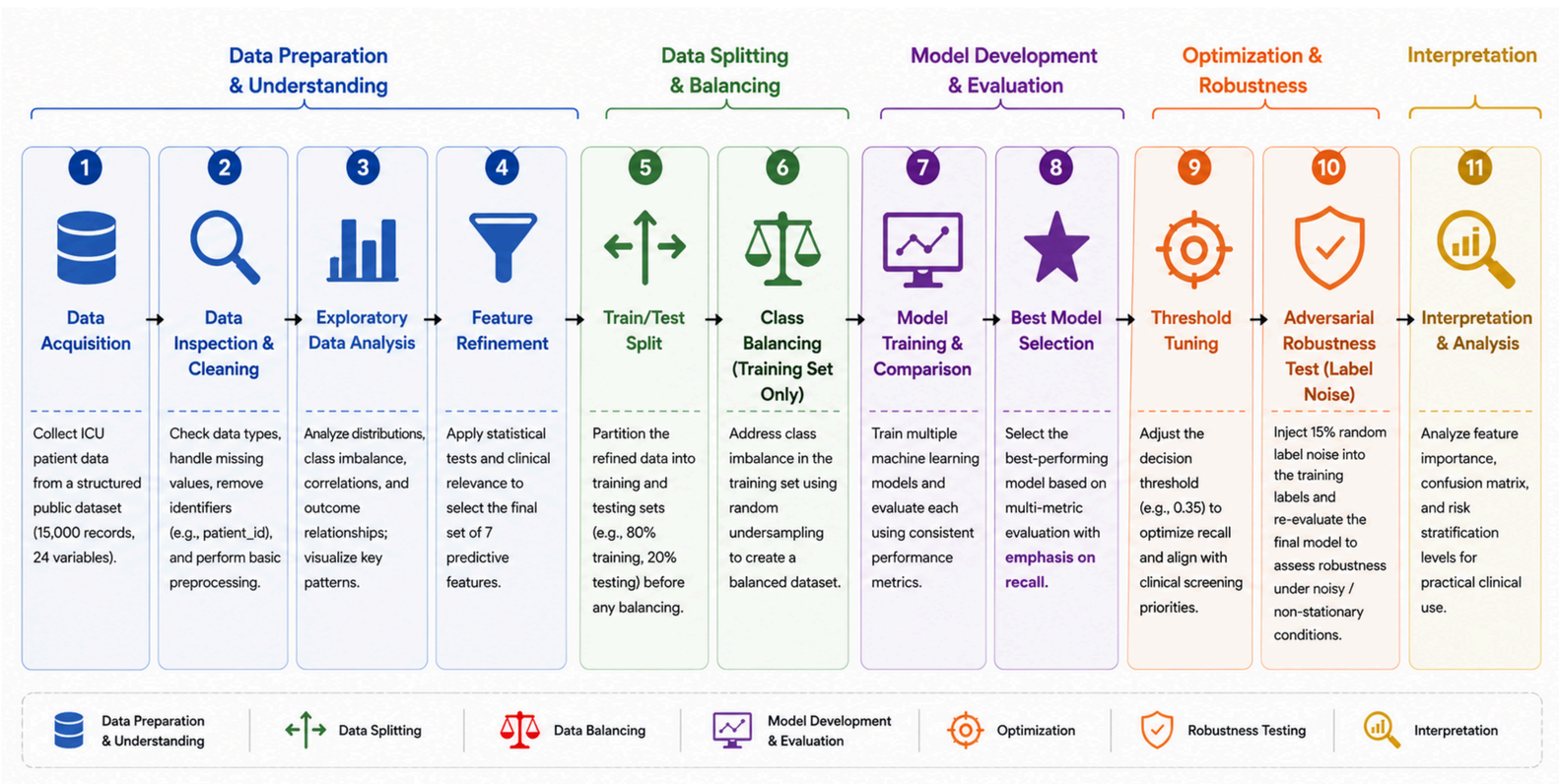
Objective

The objective of this project is to develop a practical data science framework for analyzing and predicting ICU mortality risk using structured clinical data. The approach combines data preprocessing, exploratory analysis, feature refinement, class balancing, and model evaluation within a unified workflow. Additionally, the project aims to provide interpretable results that can support real-world clinical decisions.

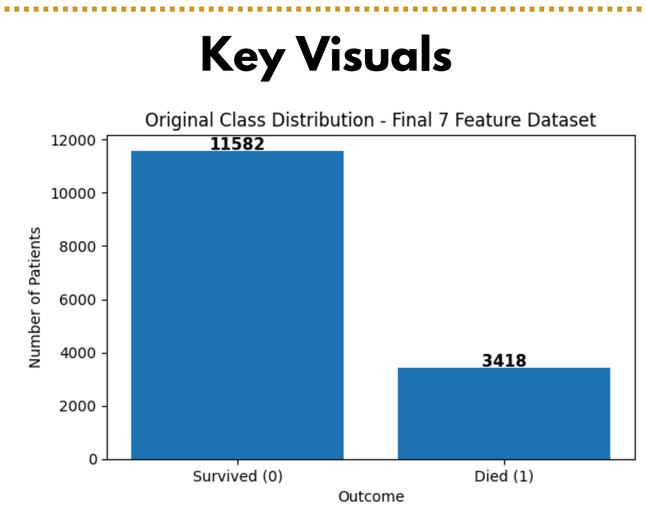
Dataset

The dataset includes 15,000 ICU patient records with 24 clinical features, reduced to 7 key predictors after analysis. The target variable is mortality (0 = survived, 1 = died). The dataset is imbalanced, with significantly more survival cases, requiring careful handling during modeling.

Methodology

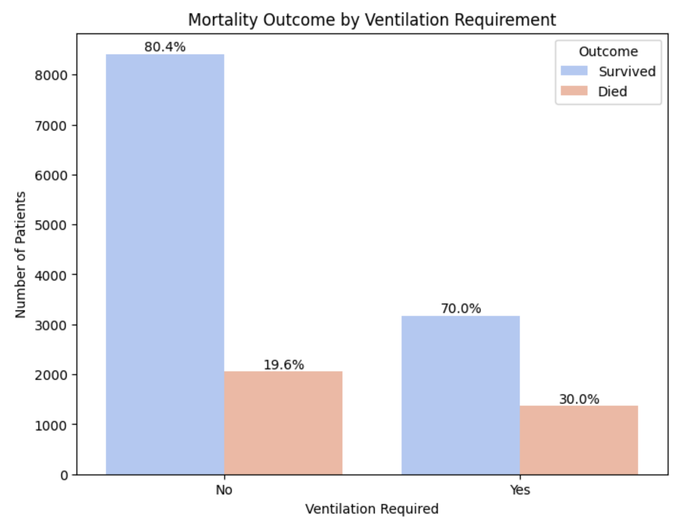


Key Visuals



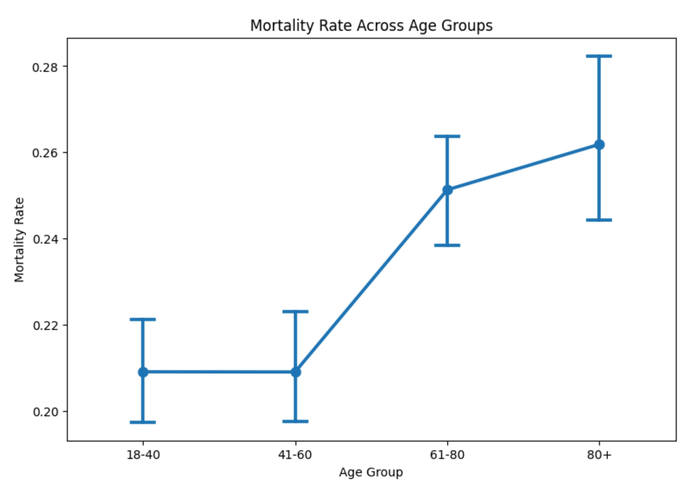
Class Distribution

This visualization highlights the severe class imbalance in the dataset, where survival cases dominate. This imbalance is a critical data challenge that influences model behavior and requires careful handling to avoid biased predictions.



Ventilation vs Mortality

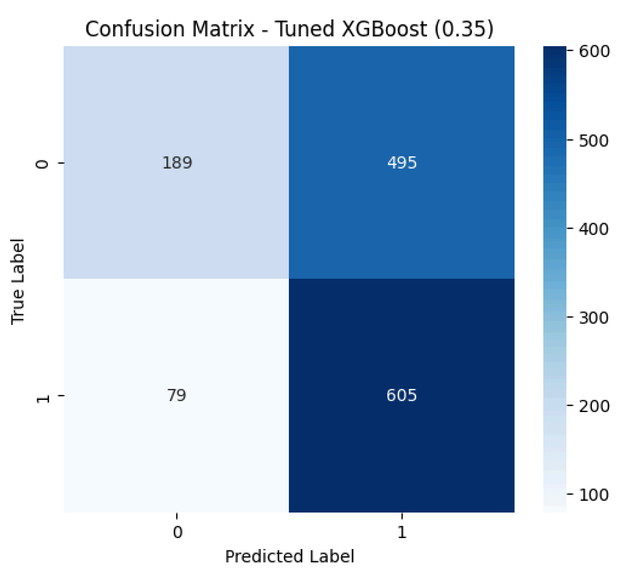
This chart shows that patients requiring ventilation have a significantly higher mortality rate, indicating that ventilation status is a strong clinical indicator of patient severity and outcome.



Mortality by Age

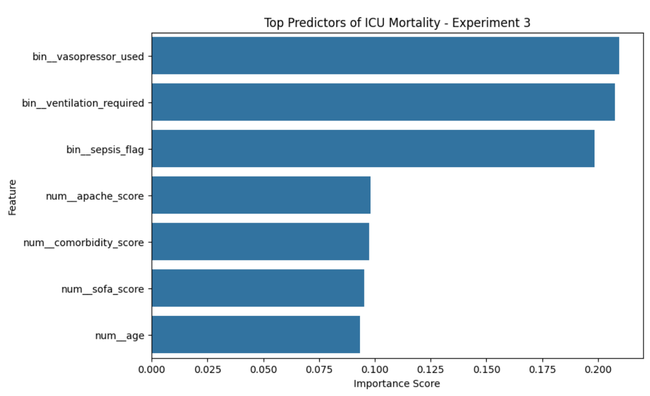
This plot demonstrates that mortality risk increases with age, especially in older groups, confirming age as an important factor in ICU outcomes.

Key Visuals



Confusion Matrix

This matrix shows that the model prioritizes identifying high-risk patients, achieving high recall while allowing some false positives, which aligns with clinical priorities.



Feature Importance

This graph identifies the most influential variables in mortality prediction, showing that clinical interventions and severity indicators contribute more strongly than demographic features.

Analysis

The results show that ICU mortality is strongly influenced by clinical severity indicators such as ventilation, vasopressor use, and sepsis. The results show that ICU mortality is strongly influenced by clinical severity indicators such as ventilation, vasopressor use, and sepsis. The trade-off between recall and precision reflects the need to align model performance with real-world clinical priorities.

Results

The results show that model performance varied across different algorithms, with ensemble methods producing the most competitive outcomes. The final selected model was a tuned XGBoost classifier with a decision threshold of 0.35, achieving a recall of 0.885, an F1-score of 0.678, and a ROC-AUC of 0.641. These findings indicate that prioritizing recall improves detection of high-risk patients, even if it introduces additional false positives. The results highlight the importance of aligning model performance with clinical priorities rather than relying on accuracy alone.

Conclusion

This study presents a practical data science framework for ICU mortality prediction that combines data analysis, feature refinement, and predictive modeling. The results show that interpretable machine learning methods can support early identification of high-risk patients while providing meaningful insights into clinical factors influencing outcomes. The study highlights that effective healthcare prediction requires balancing performance metrics, interpretability, and real-world applicability.

Limitation

- Moderate ROC-AUC performance
- Sensitive to noisy clinical data
- Based on a single dataset
- Lacks time-series patient data

Key Sources & Acknowledgements

This work is based on publicly available ICU datasets and supported by prior research in healthcare machine learning and predictive modeling. We acknowledge the contributions of existing clinical scoring systems and machine learning frameworks that informed this study.